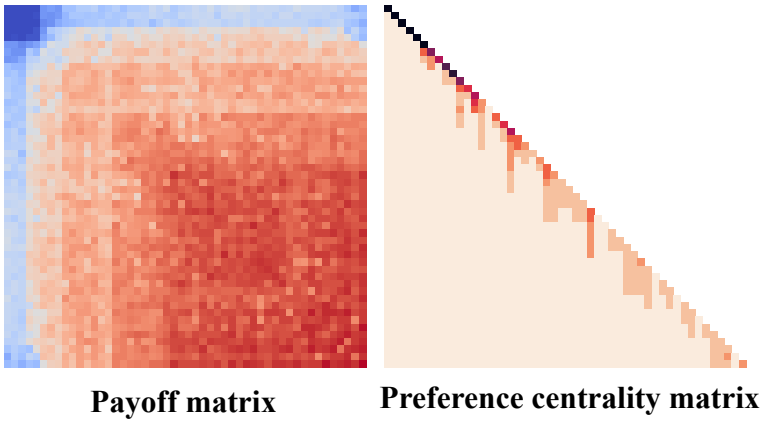

Figure 10: The learning process analysis of COLE_{SV} 1:3. A deeper shade of red in the payoff matrix signifies higher utility, while the darker-colored element on the right represents lower centrality. Clustering of darker-colored areas around the diagonal on the right indicates that the new strategy adopted in each generation is preferred by most strategies, thus overcoming the cooperative incompatibility.

one COLE_{SV} with a ratio of 0:4 or 1:3. The last column, labeled "COLEs", represents the mean rewards of the corresponding COLE_{SV} when working with other baselines. The table displays the mean cumulative rewards of each method when working with all other models in the expert group. The results indicate that COLE_{SV} 1:3 outperforms the baselines and COLE_{SV} 0:4, except in the layout of Asymmetric Advantages. In the Asymmetric Advantages, COLE_{SV} 0:4 only achieved a four-point victory over COLE_{SV} 1:3, which can be considered insignificant considering the margin of error. In the other four layouts, the rewards obtained by COLE_{SV} 1:3 while working with expert partners are significantly higher than those of COLE_{SV} 4:0 and the baselines.

Our findings indicate that COLE_{SV} 1:3 exhibits superior adaptive capacity when dealing with partners of expert levels, emphasising individual objectives is key to achieving zero-shot coordination with expert partners. To summarize, COLE_{SV} 1:3 manifests enhanced robustness and versatility in real-world environments, making it apt for collaboration with partners across a spectrum of expertise levels. Hence, in the forthcoming experiments and human-AI interaction studies, we will implement COLE_{SV} 1:3 as our definitive agent, tailored to adapt to human players of diverse skill levels.

7.3 Effectively Conquer Cooperative Incompatibility

To We analyze the learning process of COLE_{SV}, which shows that our method overcomes cooperative incompatibility.

In our analysis of the learning process of COLE_{SV} 1:3 in the Overcooked environment, as shown in Fig. 10, we observe that the method effectively overcomes the problem of cooperative incompatibility. The figure on the left in Fig. 10 shows the payoff matrix of 50 uniformly sampled checkpoints during training, with the upper left corner representing the

starting point of training. Darker red elements in the payoff matrix indicate higher rewards. The figure on the right displays the centrality matrix of preferences, which is calculated by analyzing the learning process. Unlike the payoff matrix, the darker elements in the centrality matrix indicate lower values, indicating that more strategies prefer them in the population. As shown in the figure, the darker areas cluster around the diagonal of the preference centrality matrix, indicating that most of the others prefer the updated strategy of each generation. Thus, we can conclude that our proposed COLE_{SV} effectively overcomes the problem of cooperative incompatibility.

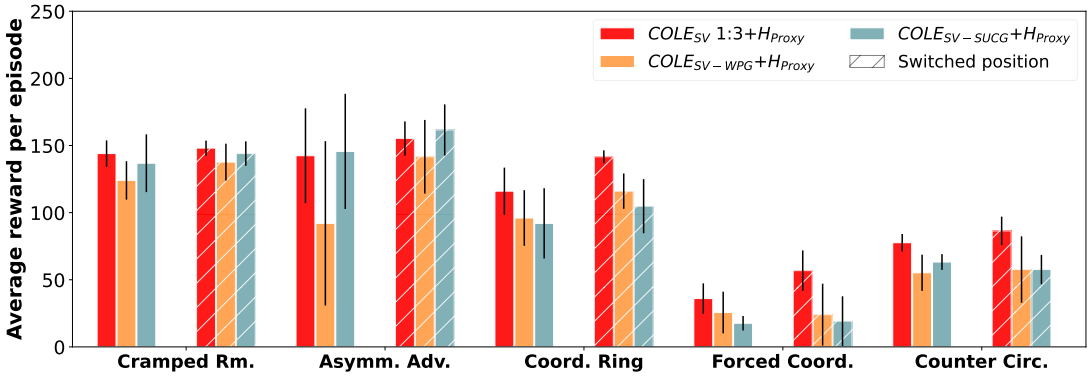

Figure 11: Performance comparison of module effectiveness in COLE_{SV} performance. The performance is presented in terms of mean episode rewards over 400 timesteps trajectories, when paired with the unseen human proxy partner H_{proxy}. $\text{COLE}_{SV\text{-}WPG}$ excludes the WPG element (Eq. 3) while computing the Shapley value. $\text{COLE}_{SV\text{-}UCGG}$ eliminates the SUCG component (Eq. 8) while sampling training partners. The results include the mean and standard error over five different random seeds. The gray bars indicate the rewards obtained when playing with themselves; the hashed bars indicate the performance when starting positions are switched.

7.4 Ablation Study

In this section, we aim to investigate the efficiency of each component within our proposed algorithm. Refer to Fig. 11 where we compare the performance of our algorithm against three ablated models namely, $\text{COLE}_{SV\text{-}WPG}$, $\text{COLE}_{SV\text{-}UCGG}$, thereby study the individual impact of these specific components. Specifically, $\text{COLE}_{SV\text{-}WPG}$ excludes the WPG element (Eq. 3) while computing the Shapley value. Therefore, in the case of $\text{COLE}_{SV\text{-}WPG}$, the computation of the coalition value employs the average of all utilities within the coalition. That is, $v(C) = \frac{1}{n^2}\sum_{i\in C}\sum_{j\in C}\mathbf{w}(i,j)$, where C represents a coalition within the full coalition set $\mathcal{N}$. Additionally, we examine the effect of the SUCG element (Eq. 8) on the performance of our proposed algorithm during the sampling of training partners. This model is referred to as $\text{COLE}_{SV\text{-}UCGG}$.

Fig. 11 illustrates the performance in terms of average episode rewards over trajectories of 400 timesteps, when coupled with the unseen human proxy partner H_{proxy}. This data

Methods	Position	Layouts				
		CRAMPED RM.	ASYMM. ADV.	COORD. RING	FORCED COORD.	COUNTER CIRC.
$\mathbf{COLE_R}$	L	131.20 (14.18)	158.40 (31.66)	80.80 (9.60)	21.60 (10.91)	63.20 (9.93)
	R	135.20 (14.40)	156.80 (19.17)	116.00 (8.39)	27.20 (26.70)	52.00 (10.73)
	AVG	133.20 (14.43)	157.60 (26.18)	98.40 (19.77)	24.40 (20.59)	57.60 (11.76)
$\mathbf{COLE_{SV}}$	L	144.00 (9.80)	142.40 (35.29)	116.00 (17.53)	36.00 (11.31)	77.60 (6.50)
	R	148.00 (5.66)	155.20 (12.75)	141.60 (4.80)	56.80 (15.05)	86.40 (10.61)
	AVG	146.00 (8.25)	148.80 (27.29)	128.80 (18.14)	46.40 (16.89)	82.00 (9.84)

Table 3: Performance comparison of COLE_{SV} and COLE_{R} on five layouts. Mean scores and standard errors (denoted in parentheses) are measured over five different seeds. In the Position column, L and R signify respective initial positions on the left and right of the layout. The term AVG stands for the average score obtained from both sides. COLE_{SV} has demonstrated significantly superior performance compared to COLE_{R} across all layouts, with the exception of the Asymm. Adv. layout. Within this specific Asymm. Adv. setting, COLE_{R} outperformed in the left position and exhibited a comparable score in the right position.

presents both the mean and the standard error across five unique random seeds. The gray bars reveal the rewards yielded when the models compete against themselves, while the hashed bars represent the performance when the starting positions are alternated. It is evident from the figure that the WPG element (Eq. 3) significantly contributes to the performance when interacting with the human proxy partner, particularly noticeable in the Asymm. Adv. layout. Furthermore, as depicted in Fig. 11, all baseline models demonstrate subpar performance when coordinating with the human proxy model in position 0 of the Asymm. Adv. layout. The computation of the Shapley value employing the WPG is fundamental to resolving the asymmetry, as depicted in Fig. 9. We think the reason behind it is that WPG can provide a just evaluation of each position's strategic cooperative ability. In relatively straightforward layouts such as Cramped Rm. and Asymm. Adv., the algorithm displays performance comparable to the ablated models $\text{COLE}_{SV\text{-}WPG}$, $\text{COLE}_{SV\text{-}UCGG}$. However, in the remaining three complex layouts, there is a significant enhancement in the COLE_{SV}'s performance.

7.5 Comparison of COLE_{SV} and COLE_{R}

Table 3 presents a comparative analysis of the coordination payoffs achieved by two practical algorithms, COLE_{SV} and COLE_{R}, across five different layouts. The results, acquired via five distinct seed values with the human proxy model H_{proxy}, feature standard errors, which are indicated within parentheses. The column named 'position' designates the diverse initial positions of the two practical algorithms respectively; here, 'L' stands for left, 'R' for right, while 'AVG' signifies the average derived from the two positions.

As demonstrated in Table 3, COLE_{SV}, utilizing the Shapley value as the core, substantially outperforms COLE_{R} in the Cramped Rm., Coord. Ring, Forced Coord., and Counter Circ. layouts. For the latter, more challenging layouts (Coord. Ring, Forced Coord., and Counter